

Experiment 4: Statistical Hypothesis Testing Using Python

Mapped CO(s): CO2

Aim: To perform statistical hypothesis testing on the Titanic dataset using Python and interpret the results using p-values.

Objectives:

- To understand the concept of hypothesis testing.
- To formulate null and alternative hypotheses.
- To perform statistical tests using Python.
- To interpret p-values and significance levels.
- To draw conclusions based on the test results.

1. Import Required Libraries

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
%matplotlib inline
```

2. Load the Titanic Dataset

The dataset is loaded using Seaborn's built-in dataset loader. If Seaborn's online dataset cannot be reached (no internet access), the dataset is loaded from a public backup CSV URL instead.

```
try:
    df = sns.load_dataset('titanic')
    print("Loaded Titanic dataset from Seaborn.")
except Exception as e:
    print("Seaborn load failed, falling back to CSV URL:", e)
    url =
    "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv"
```

```
df = pd.read_csv(url)
print("Loaded Titanic dataset from CSV URL.")

print("Shape of dataset:", df.shape)
df.head()
```

Loaded Titanic dataset from Seaborn.
Shape of dataset: (891, 15)

	survived	pclass	sex	age	sibsp	parch	fare	embarked
class \								
0	0	3	male	22.0	1	0	7.2500	S
Third								
1	1	1	female	38.0	1	0	71.2833	C
First								
2	1	3	female	26.0	0	0	7.9250	S
Third								
3	1	1	female	35.0	1	0	53.1000	S
First								
4	0	3	male	35.0	0	0	8.0500	S
Third								

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

3. Select Required Variables

For this experiment we only need the **age** and **survived** columns.

```
data = df[['age', 'survived']].copy()
data.head()
```

	age	survived
0	22.0	0
1	38.0	1
2	26.0	1
3	35.0	1
4	35.0	0

4. Remove Missing Values

```
print("Missing values before cleaning:\n", data.isnull().sum())

data_clean = data.dropna(subset=['age'])
```

```
print("\nMissing values after cleaning:\n", data_clean.isnull().sum())
print("\nShape after cleaning:", data_clean.shape)
```

Missing values before cleaning:

```
age      177
survived    0
dtype: int64
```

Missing values after cleaning:

```
age      0
survived    0
dtype: int64
```

Shape after cleaning: (714, 2)

5. Separate Passengers into Survived and Not-Survived Groups

```
survived_age = data_clean[data_clean['survived'] == 1]['age']
not_survived_age = data_clean[data_clean['survived'] == 0]['age']
```

```
print("Number of survivors (with age):", len(survived_age))
print("Number of non-survivors (with age):", len(not_survived_age))
```

```
print("\nMean age (Survived):", round(survived_age.mean(), 2))
print("Mean age (Not Survived):", round(not_survived_age.mean(), 2))
```

```
Number of survivors (with age): 290
Number of non-survivors (with age): 424
```

```
Mean age (Survived): 28.34
Mean age (Not Survived): 30.63
```

6. Formulate the Hypotheses

- **Null Hypothesis (H0):** There is no significant difference in the average age between passengers who survived and those who did not.
- **Alternative Hypothesis (H1):** There is a significant difference in the average age between passengers who survived and those who did not.

Significance level: **alpha = 0.05**

7. Perform the Independent Sample t-test

```
t_stat, p_value = stats.ttest_ind(survived_age, not_survived_age,
equal_var=False)
```

```
print("t-statistic:", round(t_stat, 4))
print("p-value:", round(p_value, 4))
```

```
t-statistic: -2.046  
p-value: 0.0412
```

8. Interpret the Result

```
alpha = 0.05  
  
if p_value < alpha:  
    print(f"p-value ({p_value:.4f}) < alpha ({alpha}) --> Reject the  
Null Hypothesis (H0).")  
    print("Conclusion: There IS a statistically significant difference  
in average age between survivors and non-survivors.")  
else:  
    print(f"p-value ({p_value:.4f}) >= alpha ({alpha}) --> Fail to  
Reject the Null Hypothesis (H0).")  
    print("Conclusion: There is NO statistically significant  
difference in average age between survivors and non-survivors.")  
  
p-value (0.0412) < alpha (0.05) --> Reject the Null Hypothesis (H0).  
Conclusion: There IS a statistically significant difference in average  
age between survivors and non-survivors.
```

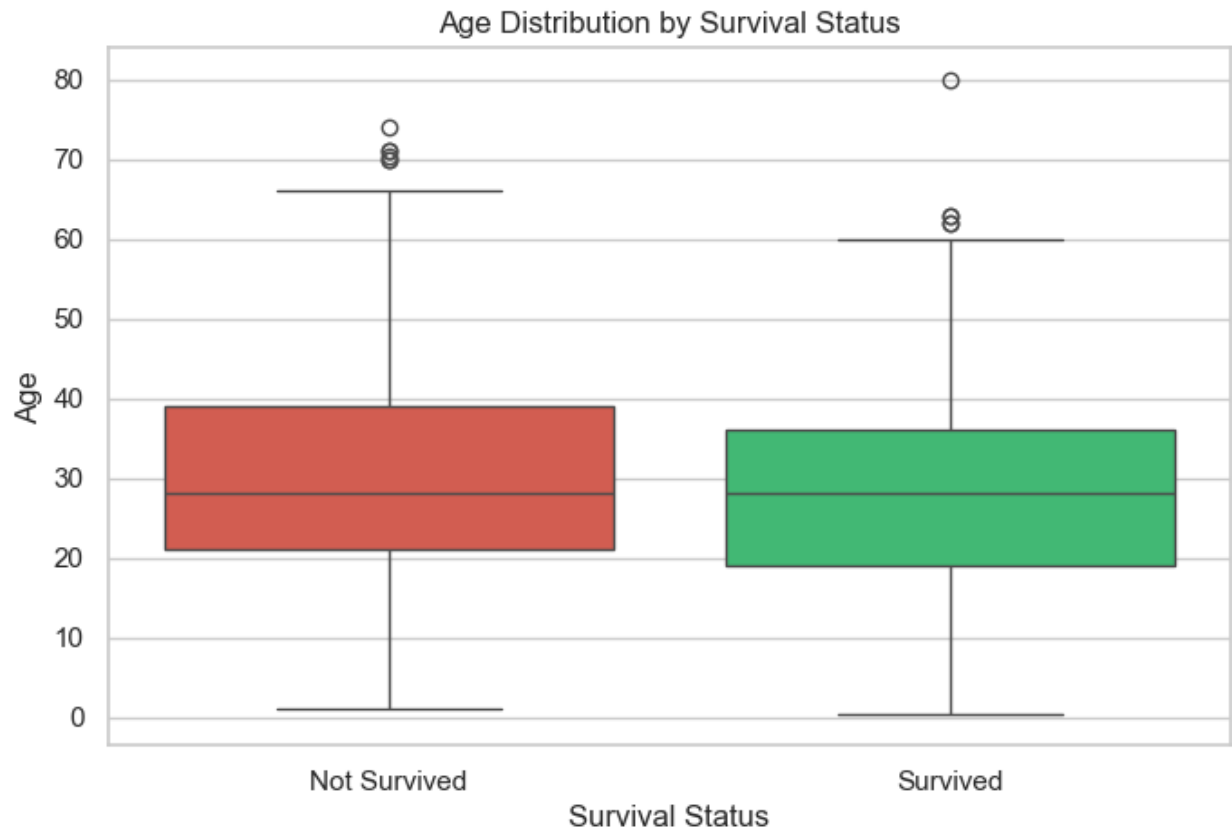
9. Visualization

```
plt.figure(figsize=(8, 5))  
sns.boxplot(x='survived', y='age', data=data_clean,  
palette=['#e74c3c', '#2ecc71'])  
plt.xticks([0, 1], ['Not Survived', 'Survived'])  
plt.xlabel('Survival Status')  
plt.ylabel('Age')  
plt.title('Age Distribution by Survival Status')  
plt.show()
```

C:\Users\choud\AppData\Local\Temp\ipykernel_1040\3388453207.py:2:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='survived', y='age', data=data_clean,  
palette=['#e74c3c', '#2ecc71'])
```



```
plt.figure(figsize=(8, 5))
sns.histplot(survived_age, color='#2ecc71', label='Survived',
kde=True, alpha=0.5)
sns.histplot(not_survived_age, color='#e74c3c', label='Not Survived',
kde=True, alpha=0.5)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Age Distribution: Survived vs Not Survived')
plt.legend()
plt.show()
```



Result

Statistical hypothesis testing was successfully performed using Python. The hypothesis was tested using an Independent Sample t-test, and the conclusion was drawn based on the calculated p-value.

Conclusion

The experiment demonstrated the application of statistical hypothesis testing using Python. The Independent Sample t-test was used to compare the mean age of survived and non-survived passengers. Based on the p-value, an appropriate statistical conclusion was made regarding the null hypothesis. Hypothesis testing is an important tool for making data-driven decisions and validating assumptions in data science.

Exercises / Assignments

Exercise 1: Perform a t-test using the Fare variable instead of Age

H0: There is no significant difference in the average fare between passengers who survived and those who did not. **H1:** There is a significant difference in the average fare between passengers who survived and those who did not.

```
fare_data = df[['fare', 'survived']].dropna()

survived_fare = fare_data[fare_data['survived'] == 1]['fare']
not_survived_fare = fare_data[fare_data['survived'] == 0]['fare']

t_stat_fare, p_value_fare = stats.ttest_ind(survived_fare,
not_survived_fare, equal_var=False)

print("Mean Fare (Survived):", round(survived_fare.mean(), 2))
print("Mean Fare (Not Survived):", round(not_survived_fare.mean(), 2))
print("t-statistic:", round(t_stat_fare, 4))
print("p-value:", round(p_value_fare, 4))

if p_value_fare < 0.05:
    print("\nResult: Reject H0 -> Significant difference in average
fare between the two groups.")
else:
    print("\nResult: Fail to reject H0 -> No significant difference in
average fare between the two groups.")

Mean Fare (Survived): 48.4
Mean Fare (Not Survived): 22.12
t-statistic: 6.8391
p-value: 0.0

Result: Reject H0 -> Significant difference in average fare between
the two groups.
```

Exercise 2: Change the significance level from 0.05 to 0.01 and compare the result

Re-evaluate the original Age t-test result at a stricter significance level ($\alpha = 0.01$).

```
alpha_01 = 0.01

print("Using the Age t-test results: t-statistic =", round(t_stat, 4),
", p-value =", round(p_value, 4))
print()
```

```

print("At alpha = 0.05:")
print("  ->", "Reject H0" if p_value < 0.05 else "Fail to Reject H0")

print("\nAt alpha = 0.01:")
print("  ->", "Reject H0" if p_value < alpha_01 else "Fail to Reject H0")

print("\nComparison: A stricter significance level (0.01) requires stronger evidence before rejecting H0.")
print("If the p-value lies between 0.01 and 0.05, the decision changes between the two significance levels;")
print("otherwise, the conclusion remains the same at both levels.")

```

Using the Age t-test results: t-statistic = -2.046 , p-value = 0.0412

At alpha = 0.05:

-> Reject H0

At alpha = 0.01:

-> Fail to Reject H0

Comparison: A stricter significance level (0.01) requires stronger evidence before rejecting H0.

If the p-value lies between 0.01 and 0.05, the decision changes between the two significance levels;
otherwise, the conclusion remains the same at both levels.

Exercise 3: Interpret the obtained p-value

A short written interpretation of the p-values obtained above (Age and Fare tests).

```

print("Interpretation:")
print(f"- Age test: p-value = {p_value:.4f}. This is the probability of observing a difference in mean")
print("  age as extreme as (or more extreme than) the one observed, assuming H0 (no difference) is true.")
print(f"  Since p-value {'<' if p_value < 0.05 else '>='} 0.05, we", "reject" if p_value < 0.05 else "fail to reject", "H0 for the Age variable.")
print()
print(f"- Fare test: p-value = {p_value_fare:.6f}. Since p-value {'<' if p_value_fare < 0.05 else '>='} 0.05,")
print("  we", "reject" if p_value_fare < 0.05 else "fail to reject", "H0 for the Fare variable.")
print()
print("A smaller p-value indicates stronger evidence against the null hypothesis, i.e. stronger evidence")

```



```
print("that the observed difference between groups is unlikely to have occurred by random chance alone.")
```

Interpretation:

- Age test: p-value = 0.0412. This is the probability of observing a difference in mean age as extreme as (or more extreme than) the one observed, assuming H_0 (no difference) is true.
Since p-value < 0.05, we reject H_0 for the Age variable.
- Fare test: p-value = 0.000000. Since p-value < 0.05, we reject H_0 for the Fare variable.

A smaller p-value indicates stronger evidence against the null hypothesis, i.e. stronger evidence that the observed difference between groups is unlikely to have occurred by random chance alone.

Exercise 4: Apply another statistical test such as the Chi-Square test

Here we test whether **Passenger Class (pclass)** and **Survival (survived)** are independent of each other.

H_0 : Passenger class and survival status are independent (no association). **H_1 :** Passenger class and survival status are associated (not independent).

```
contingency_table = pd.crosstab(df['pclass'], df['survived'])
print("Contingency Table (Pclass vs Survived):")
print(contingency_table)

chi2_stat, p_value_chi2, dof, expected =
stats.chi2_contingency(contingency_table)

print("\nChi-Square Statistic:", round(chi2_stat, 4))
print("Degrees of Freedom:", dof)
print("p-value:", round(p_value_chi2, 6))

if p_value_chi2 < 0.05:
    print("\nResult: Reject  $H_0$  -> Passenger class and survival are significantly associated.")
else:
    print("\nResult: Fail to reject  $H_0$  -> No significant association between passenger class and survival.")

Contingency Table (Pclass vs Survived):
survived    0    1
pclass
1           80  136
```

2	97	87
3	372	119

Chi-Square Statistic: 102.889
Degrees of Freedom: 2
p-value: 0.0

Result: Reject H0 -> Passenger class and survival are significantly associated.

Exercise 5: State the null and alternative hypotheses for a different variable

Variable chosen: Sex (Gender) vs Survival

- **Null Hypothesis (H0):** There is no significant association between a passenger's gender and their survival status (gender and survival are independent).
- **Alternative Hypothesis (H1):** There is a significant association between a passenger's gender and their survival status (gender and survival are not independent).

This can be tested using a Chi-Square test of independence, as shown below (for verification).

```
contingency_sex = pd.crosstab(df['sex'], df['survived'])
print("Contingency Table (Sex vs Survived):")
print(contingency_sex)

chi2_sex, p_value_sex, dof_sex, expected_sex =
stats.chi2_contingency(contingency_sex)

print("\nChi-Square Statistic:", round(chi2_sex, 4))
print("p-value:", round(p_value_sex, 6))

if p_value_sex < 0.05:
    print("\nResult: Reject H0 -> Gender and survival status are
significantly associated.")
else:
    print("\nResult: Fail to reject H0 -> No significant association
between gender and survival status.")

Contingency Table (Sex vs Survived):
survived    0    1
sex
female      81  233
male       468  109

Chi-Square Statistic: 260.717
p-value: 0.0
```

Result: Reject H_0 -> Gender and survival status are significantly associated.